MAT 3007 – Optimization

Linear Optimization and Graphical Solutions

Lecture 05

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CUHK-Shenzhen

Logistics & Content

The deadline of the first assignment is extended to next Tuesday, Feb 18th, 11 p.m.

Today's topics:

- Further modeling techniques and examples.
- Graphical solutions.

Recap: Linear Problems

Linear Optimization Problem:

$$
\begin{aligned}
\text{minimize/maximize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\
\text{subject to} \quad & \boldsymbol{A}_1\boldsymbol{x} \geq \boldsymbol{b} \\
& \boldsymbol{A}_2\boldsymbol{x} \leq \boldsymbol{d} \\
& \boldsymbol{A}_3\boldsymbol{x} = \boldsymbol{e} \\
& x_i \geq 0 \quad \forall\ i \in N_1 \\
& x_i \leq 0 \quad \forall\ i \in N_2 \\
& x_i \text{ free} \quad \forall\ i \in N_3
\end{aligned}
$$

Standard Form of an LP:

$$
\begin{aligned}
\text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\
\text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\
& \boldsymbol{x} \geq \boldsymbol{0}
\end{aligned}
$$

A General Modeling Tool

Consider the optimization problems:

$$\text{minimize}_{\boldsymbol{x}} \ \sum_{i=1}^{n} f_i(\boldsymbol{x}) \quad \text{s.t.} \quad \boldsymbol{x} \in \Omega \tag{1}$$

$$\text{minimize}_{\boldsymbol{x},\boldsymbol{t}} \ \sum_{i=1}^{n} t_i \quad \text{s.t.} \quad \boldsymbol{x} \in \Omega, \quad f_i(\boldsymbol{x}) \leq t_i, \quad \forall\ i \tag{2}$$

where $f_i : \mathbb{R}^n \to \mathbb{R}$ are given and $\Omega \subset \mathbb{R}^n$ is the feasible set.

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where $f_i : \mathbb{R}^n \to \mathbb{R}$ are given and $\Omega \subset \mathbb{R}^n$ is the feasible set.

Lemma: Modeling Tool

The problems (1) and (2) are equivalent in the following way:

- If $\boldsymbol{x}^*$ is an optimal sol. of (1), then $(\boldsymbol{x}^*, t_1^*, \ldots, t_n^*))$ with $t_i^* = f_i(\boldsymbol{x}^*)$ is an optimal sol. of (2).
- If $(\boldsymbol{x}^*, t_1^*, \ldots, t_n^*)$ is an optimal sol. of (2), then $\boldsymbol{x}^*$ is an optimal sol. of (1).

Also, both problems have the same optimal value.

Proof sketch: We assume $n = 1$ for simplicity.

(a) Assume $\boldsymbol{x}^* \in \Omega$ is an optimal sol. to (1). Then $(\boldsymbol{x}^*, t^*)$ with $t^* = f(\boldsymbol{x}^*)$ is feasible to (2).

Proof sketch: We assume $n = 1$ for simplicity.
(a) Assume $\boldsymbol{x}^* \in \Omega$ is an optimal sol. to (1). Then $(\boldsymbol{x}^*, t^*)$ with $t^* = f(\boldsymbol{x}^*)$ is feasible to (2). Also, since $\boldsymbol{x}^*$ is optimal, we have $f(\boldsymbol{x}) \geq f(\boldsymbol{x}^*),\ \forall \boldsymbol{x} \in \Omega$.

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$$t \geq f(\boldsymbol{x}) \geq f(\boldsymbol{x}^*) = t^*.$$

Therefore, $(\boldsymbol{x}^*, t^*)$ is optimal for (2),

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(b) Assume $(\boldsymbol{x}^*, t^*)$ is optimal for (2). By feasibility, $x^* \in \Omega$, so x^* is feasible for (1).

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(b) Assume $(\boldsymbol{x}^*, t^*)$ is optimal for (2). By feasibility, $x^* \in \Omega$, so x^* is feasible for (1). Also, observe that for any $\boldsymbol{x} \in \Omega$, $(\boldsymbol{x}, t)$ with $t = f(\boldsymbol{x})$ is feasible for (2). Therfore, by optimality of $(\boldsymbol{x}^*, t^*)$ for (2), we have $f(\boldsymbol{x}) = t \geq t^*$.

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(a) Assume $\boldsymbol{x}^* \in \Omega$ is an optimal sol. to (1). Then $(\boldsymbol{x}^*, t^*)$ with $t^* = f(\boldsymbol{x}^*)$ is feasible to (2). Also, since $\boldsymbol{x}^*$ is optimal, we have $f(\boldsymbol{x}) \geq f(\boldsymbol{x}^*),\ \forall \boldsymbol{x} \in \Omega$. For any feasible point $(\boldsymbol{x}, t)$ of (2),

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$$f(\boldsymbol{x}) = t \geq t^* \geq f(\boldsymbol{x}^*), \forall x \in \Omega.$$

Therefore, $\boldsymbol{x}^*$ is optimal for (1).

Take Another Look

We can think about this equivalency in another way:

$$\text{minimize}_{\boldsymbol{x}} \sum_{i=1}^{n} f_i(\boldsymbol{x}) \quad \text{s.t.} \quad \boldsymbol{x} \in \Omega$$

is equivalent to

$$\text{minimize}_{\boldsymbol{x},\boldsymbol{t}} \sum_{i=1}^{n} t_i \quad \text{s.t.} \quad \boldsymbol{x} \in \Omega, \quad f_i(\boldsymbol{x}) = t_i, \quad \forall\ i,$$

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which is further equivalent to

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Why? Intuition: since we minimize $\sum_i t_i$ in (3), it pushes each t_i to take the value of $f_i(\boldsymbol{x})$.

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Why? Intuition: since we minimize $\sum_i t_i$ in (3), it pushes each t_i to take the value of $f_i(\boldsymbol{x})$.

- Proof idea: if $(\boldsymbol{x}^*, \boldsymbol{t}^*)$ is an optimal sol. to (3) and $f_i(\boldsymbol{x}^*) < t_i^*$ for some i, then we can find another feasible solution with smaller objective value $\rightsquigarrow$ contradiction! (See Lecture 3)

A General Modeling Tool – Maximization

Consider the optimization problems:

$$\text{maximize}_{\boldsymbol{x}} \ \sum_{i=1}^{n} f_i(\boldsymbol{x}) \quad \text{s.t.} \quad \boldsymbol{x} \in \Omega \tag{4}$$

$$\text{maximize}_{\boldsymbol{x},\boldsymbol{t}} \ \sum_{i=1}^{n} t_i \quad \text{s.t.} \quad \boldsymbol{x} \in \Omega, \quad f_i(\boldsymbol{x}) \geq t_i, \quad \forall\ i \tag{5}$$

where $f_i : \mathbb{R}^n \to \mathbb{R}$ are given and $\Omega \subset \mathbb{R}^n$ is the feasible set.

A General Modeling Tool – Maximization

Consider the optimization problems:

$$\text{maximize}_{\boldsymbol{x}} \ \sum_{i=1}^{n} f_i(\boldsymbol{x}) \quad \text{s.t.} \quad \boldsymbol{x} \in \Omega \tag{4}$$

$$\text{maximize}_{\boldsymbol{x},\boldsymbol{t}} \ \sum_{i=1}^{n} t_i \quad \text{s.t.} \quad \boldsymbol{x} \in \Omega, \quad f_i(\boldsymbol{x}) \geq t_i, \quad \forall\ i \tag{5}$$

where $f_i : \mathbb{R}^n \to \mathbb{R}$ are given and $\Omega \subset \mathbb{R}^n$ is the feasible set.

Lemma: Modeling Tool

The problems (4) and (5) are equivalent in the following way:

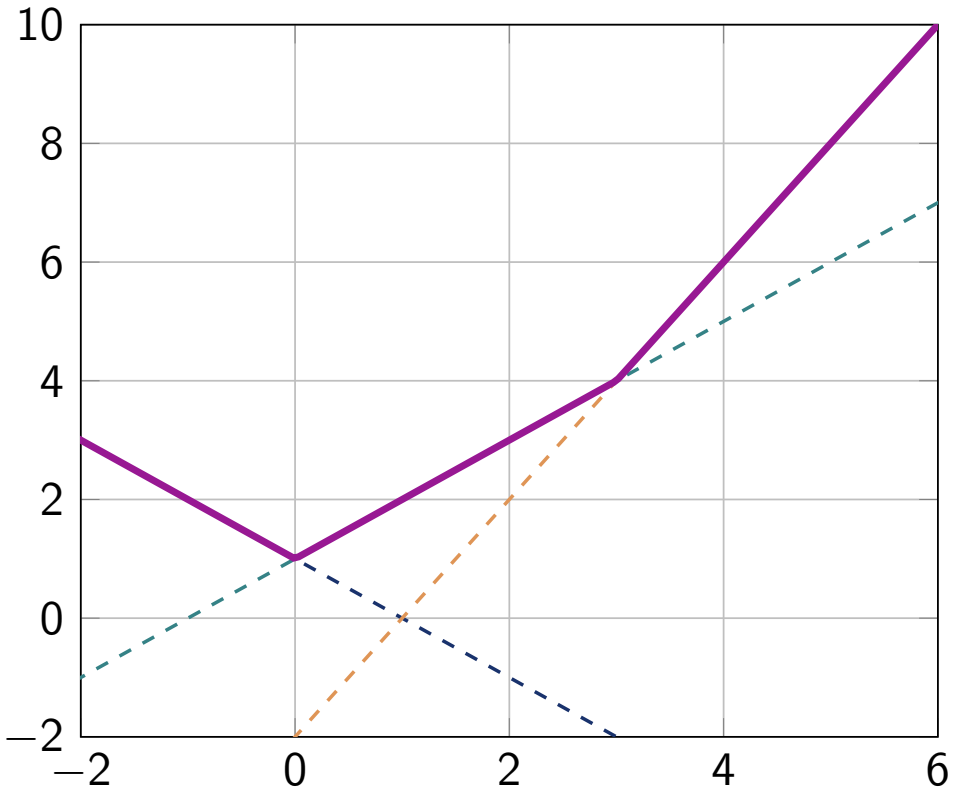
- If $\boldsymbol{x}^*$ is an optimal sol. of (4), then $(\boldsymbol{x}^*, t_1^*, \ldots, t_n^*))$ with $t_i^* = f_i(\boldsymbol{x}^*)$ is an optimal sol. of (5).
- If $(\boldsymbol{x}^*, t_1^*, \ldots, t_n^*)$ is an optimal sol. of (5), then $\boldsymbol{x}^*$ is an optimal sol. of (4).

Also, both problems have the same optimal value.

Example: Minimax Objective

We are interested in the minimax objective:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}} \quad & \max_{i=1,\ldots,n}\{\boldsymbol{c}_i^\top \boldsymbol{x} + d_i\} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0}. \end{aligned}$$



- Plot of the function: $\max\{1-x, x+1, 2x-2\}$

Reformulating Minimax Objectives

1. Introduce an auxiliary variable:
   - Define $y = \max_{i=1,\ldots,n}\{\boldsymbol{c}_i^\top \boldsymbol{x} + d_i\}$.

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$$\begin{aligned} \text{minimize}_{\boldsymbol{x},y} \quad & y \\ \text{subject to} \quad & \max_i\{\boldsymbol{c}_i^\top \boldsymbol{x} + d_i\} \le \boldsymbol{y} \\ & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \qquad \boldsymbol{x} \ge \boldsymbol{0}. \end{aligned}$$

Reformulating Minimax Objectives

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3. Final reformulation and equivalences:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x},y} \quad & y \\ \text{subject to} \quad & \boldsymbol{c}_i^\top \boldsymbol{x} + d_i \le y \quad \forall\ i, \\ & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \qquad \boldsymbol{x} \ge \boldsymbol{0}. \end{aligned}$$

Example: Absolute Values

Problems with absolute values might be handled as well by LPs:

$$\begin{aligned} \text{minimize} \quad & \textstyle\sum_{i=1}^{n} |x_i| \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}. \end{aligned}$$

This can be equivalently written as

$$\begin{aligned} \text{minimize}_{\boldsymbol{x},\boldsymbol{t}} \quad & \textstyle\sum_{i=1}^{n} t_i \\ \text{s.t.} \quad & t_i \geq x_i \\ & t_i \geq -x_i \\ & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}. \end{aligned}$$

$t_i \geq |x_i|$

$\Leftrightarrow \begin{cases} t_i \geq x_i \\ t_i \geq -x_i \end{cases}$

- Similar ideas can be applied for constraints like $|\boldsymbol{a}^\top \boldsymbol{x} + b| \leq c$.

Absolute Values: An Observation

Consider a similar problem:

$$\begin{aligned} \text{maximize} \quad & \sum_{i=1}^{n} |x_i| \\ \text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}. \end{aligned}$$

Can we use the similar idea and transform it into:

$$\begin{aligned} \text{maximize} \quad & \sum_{i=1}^{n} t_i \\ \text{s.t.} \quad & t_i \geq x_i \\ & t_i \geq -x_i \\ & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}. \end{aligned}$$

- **Answer: No.**

Recap: Linear Algebra

- A short note about linear algebra was posted on BB.

Linear Combination and Span

The linear combination of a set of vectors $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_m$ in $\mathbb{R}^n$ with coefficient $a_1, \cdots, a_m \in \mathbb{R}$ is

$$a_1\boldsymbol{x}_1 + a_2\boldsymbol{x}_2 + \cdots + a_m\boldsymbol{x}_m.$$

The span of this set of vectors is set of all linear combinations:

$$span(\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_m) := \{a_1\boldsymbol{x}_1 + a_2\boldsymbol{x}_2 + \cdots + a_m\boldsymbol{x}_m : a_1, \cdots, a_m \in \mathbb{R}\}.$$

Linear Combination and Linear Dependence

Linear Combination and Span

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Linear Dependence

A set of vectors $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_m$ in $\mathbb{R}^n$ is linearly dependent if there exist scalars $a_1, \cdots, a_m \in \mathbb{R}$, not all zero, such that $\sum_{i=1}^m a_i\boldsymbol{x}_i = \boldsymbol{0}$. We call them linearly independent is they are not linearly dependent.

Basis

A set of vectors $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_m\}$ in V is called a **basis** of a vector space V if:

- The vectors are linearly independent, and
- They span V, meaning any vector in V can be expressed as a linear combination of $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_m$.

Basis, Kernel and Range

Basis

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Kernel and Range of a Matrix A

Kernel (or Null Space) of $\boldsymbol{A}$: The set of all vectors $\boldsymbol{x}$ such that $\boldsymbol{Ax} = \boldsymbol{0}$.

Range (or image) of $\boldsymbol{A}$: The set of all vectors $\boldsymbol{b}$ such that $\boldsymbol{Ax} = \boldsymbol{b}$ for some $\boldsymbol{x}$. It is the span of the columns of $\boldsymbol{A}$.

Rank of a Matrix A

The **rank** of a matrix $\boldsymbol{A}$ is the dimension of the **range** of $\boldsymbol{A}$. It is equal to the number of linearly independent columns (or rows).

$$\text{rank}(A) = \dim(\text{range}(A)).$$

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Full Column-Rank, Full Row-Rank, and Full-Rank

Let A be an $m \times n$ matrix.

- A has **full column-rank** if its rank is equal to the number of columns: $\text{rank}(A) = n$. This means all columns of A are linearly independent.
- A has **full row-rank** if its rank is equal to the number of rows: $\text{rank}(A) = m$. This means all rows of A are linearly independent.
- A is **full-rank** if its rank is equal to the largest possible rank for a matrix of its size: $\text{rank}(A) = \min(m, n)$. This means A has either full column-rank, full row-rank, or both.

Solving LPs Graphically

Starting Point: Graphical Solutions to LP

It is very helpful to study a small LP from a graphical point of view.

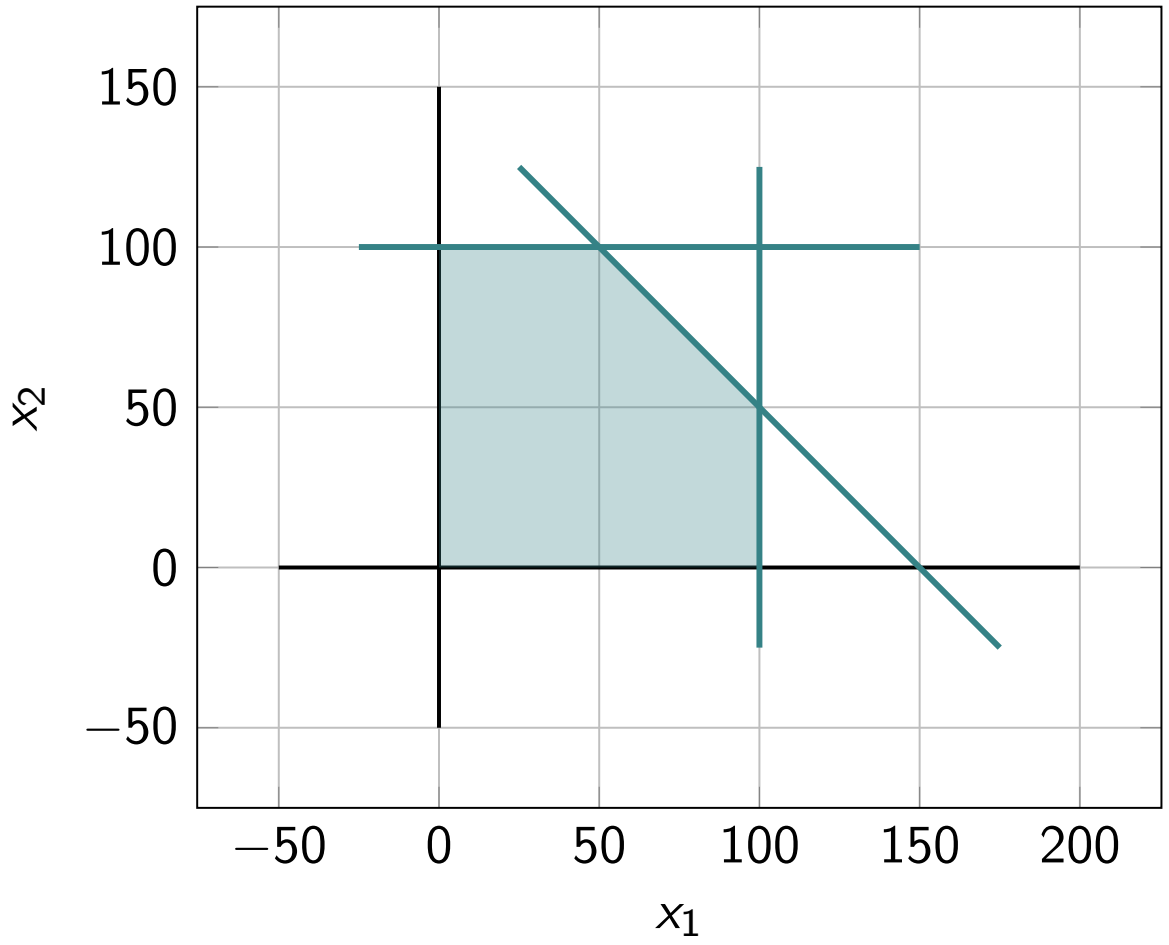
Recall the production problem:

$$
\begin{array}{rrrl}
\text{maximize} & x_1 & +2x_2 & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0.
\end{array}
$$

How can we solve this using a graph?

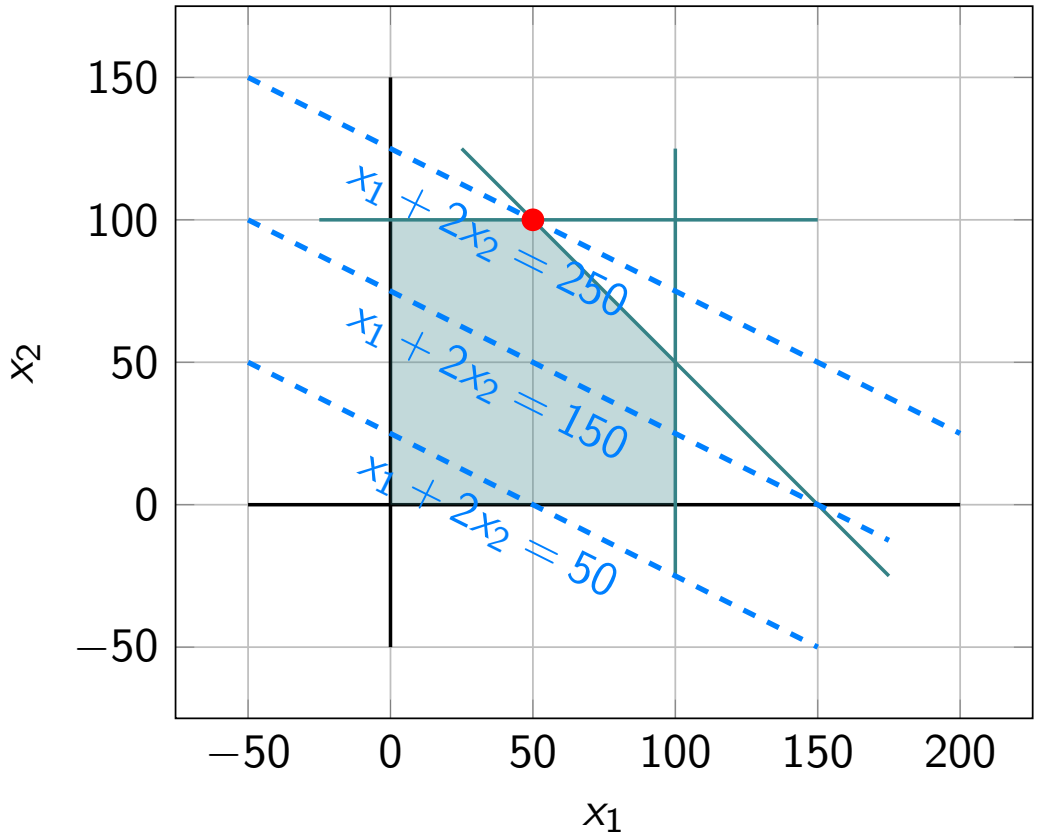
Solve LP from Graph

We first draw the feasible region.



To Maximize $x_1 + 2x_2$ ...

We then draw the function $x_1 + 2x_2 = c$ for different values of c.



- The optimal solution is the highest one among these lines that touches the feasible region.
- **Optimal solution**: $(50, 100)$. **Objective value**: 250.

Solving Another Simple LP

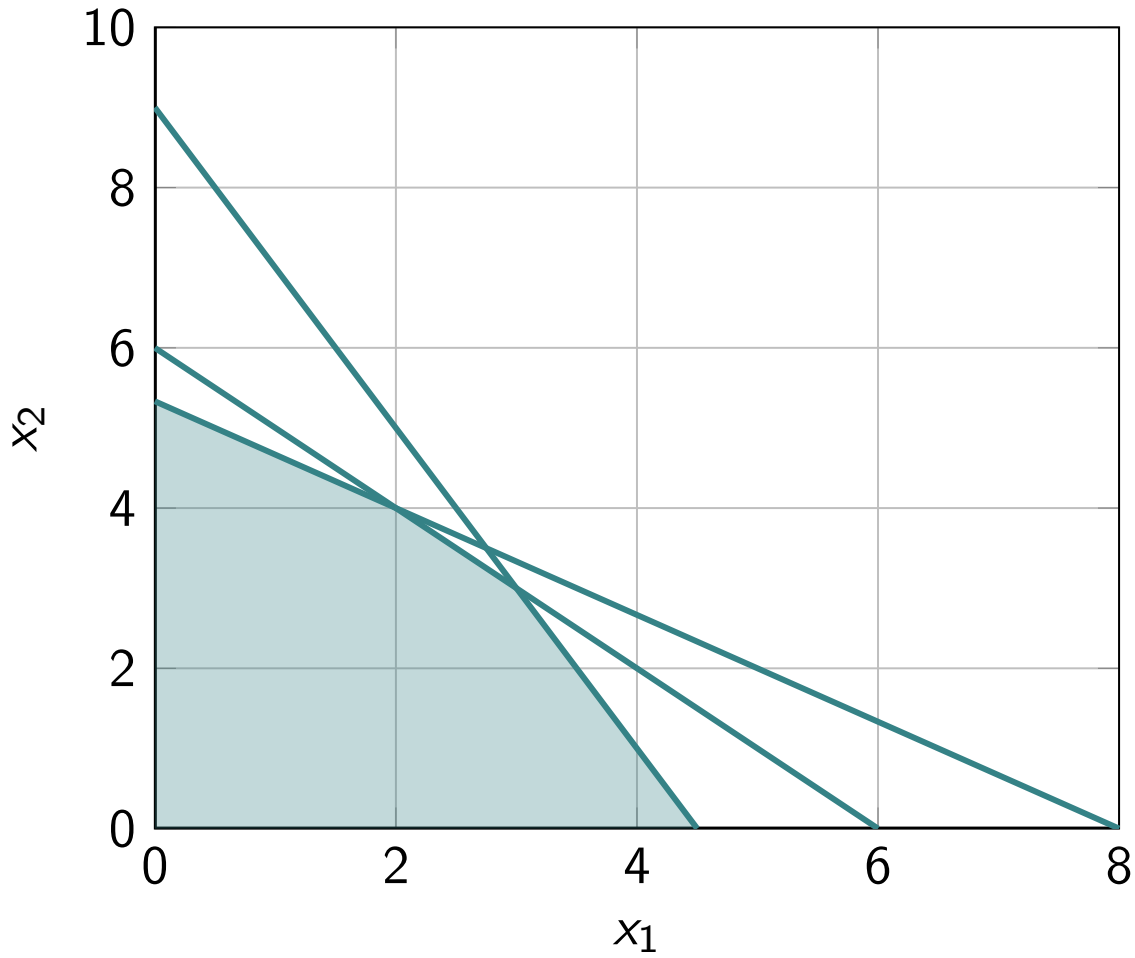
Before introducing more formal definitions and concepts, let us consider a second example:

$$
\begin{array}{rrrl}
\text{maximize} & 6x_1 & +4x_2 & \\
\text{subject to} & x_1 & +x_2 & \leq 6 \\
& 2x_1 & x_2 & \leq 9 \\
& 2x_1 & +3x_2 & \leq 16 \\
& x_1, & x_2 & \geq 0.
\end{array}
$$

How can we solve this using a graph?

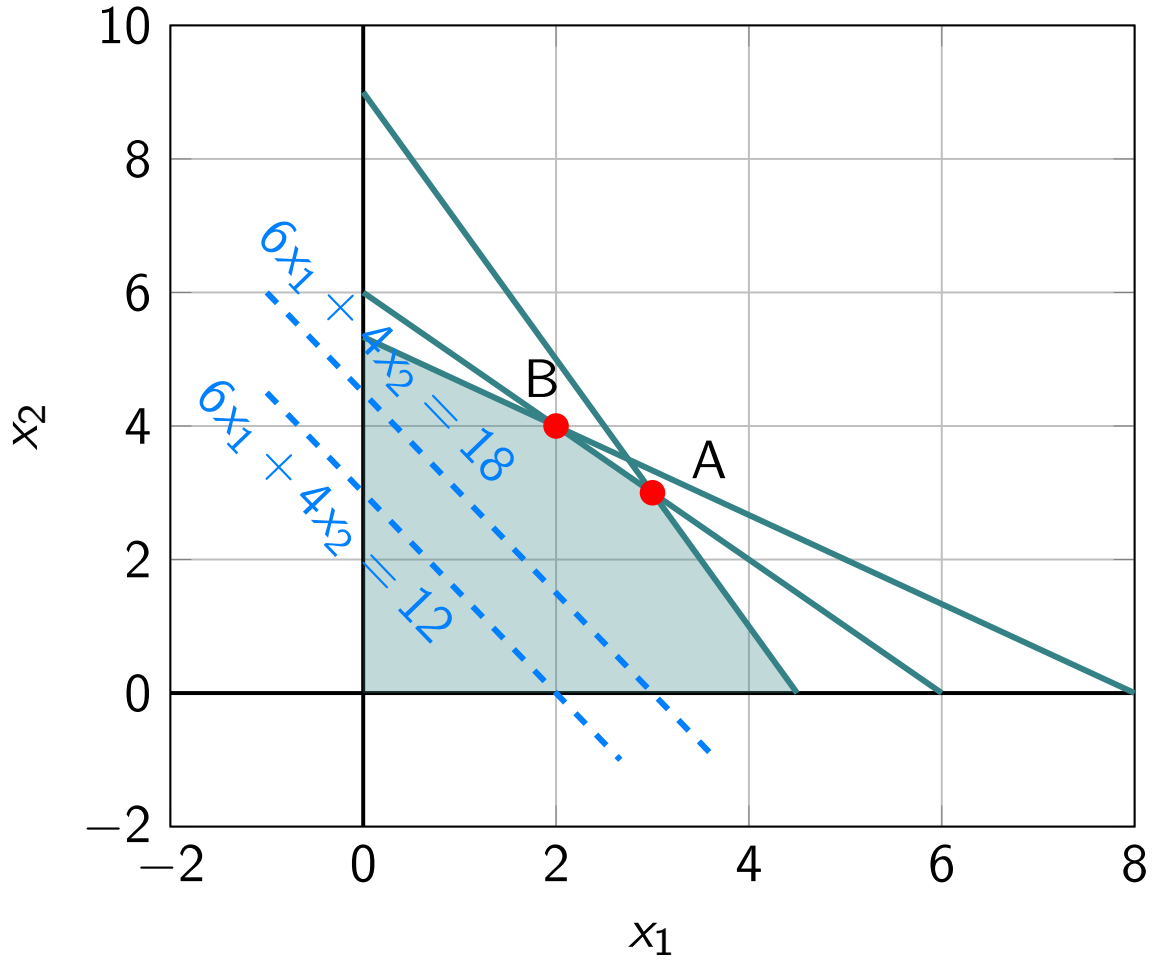
Solve LP from Graph

We again first draw the feasible region.



To Maximize $6x_1 + 4x_2$ ...

We again draw the contour lines $6x_1 + 4x_2 = c$ for different c.



- Is $A = (3, 3)$ or $B = (2, 4)$ the optimal solution?

Some Observations

- The feasible region of an LP is a polyhedron.
- The optimal solution tends to be a corner of the feasible region.
- Some constraints are active at the optimal solution ($x_2 \leq 100$, $x_1 + x_2 \leq 150$), some are not ($x_1 < 100$).

Next Goal: We will formalize these observations and study algorithms for solving LPs that can:

- Guarantee to find the optimal solution.
- Run within a certain (reasonable) amount of time.

Some Definitions: Polyhedron

Polyhedron

A polyhedron is a set that can be written in the form:

$$\{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{A}\boldsymbol{x} \geq \boldsymbol{b}\},$$

where $\boldsymbol{A}$ is an $m \times n$ matrix and $\boldsymbol{b} \in \mathbb{R}^m$.

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where $\boldsymbol{A}$ is an $m \times n$ matrix and $\boldsymbol{b} \in \mathbb{R}^m$.

- Recall that in the standard form of LP, the feasible set is

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{x} \geq \boldsymbol{0}.$$

- Is this a polyhedron? Why?

Some Definitions: Polyhedron

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- Recall that in the standard form of LP, the feasible set is

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \quad \boldsymbol{x} \geq \boldsymbol{0}.$$

- Is this a polyhedron? Why?
- Yes, we can write it as $\boldsymbol{A}\boldsymbol{x} \geq \boldsymbol{b}$, $\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$, $\boldsymbol{I} \cdot \boldsymbol{x} \geq \boldsymbol{0}$ where $\boldsymbol{I}$ is the identity matrix.

Convex Sets and Convex Combinations

Convex Combination

For any $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$ and $\lambda_1, \ldots, \lambda_n \geq 0$ satisfying $\lambda_1 + \cdots + \lambda_n = 1$, we call $\sum_{i=1}^n \lambda_i \boldsymbol{x}_i$ a convex combination of $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n$.

Convex Sets and Convex Combinations

Convex Combination

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Definition: Convex Set

A set $S \subseteq \mathbb{R}^n$ is convex if for any $\boldsymbol{x}$, $\boldsymbol{y} \in S$, and any $\lambda \in [0, 1]$, $\lambda \boldsymbol{x} + (1 - \lambda)\boldsymbol{y} \in S$.

- Intuitively, this means that the line segment between two points in the set must also lie in the set.



convex



nonconvex

Extreme Points

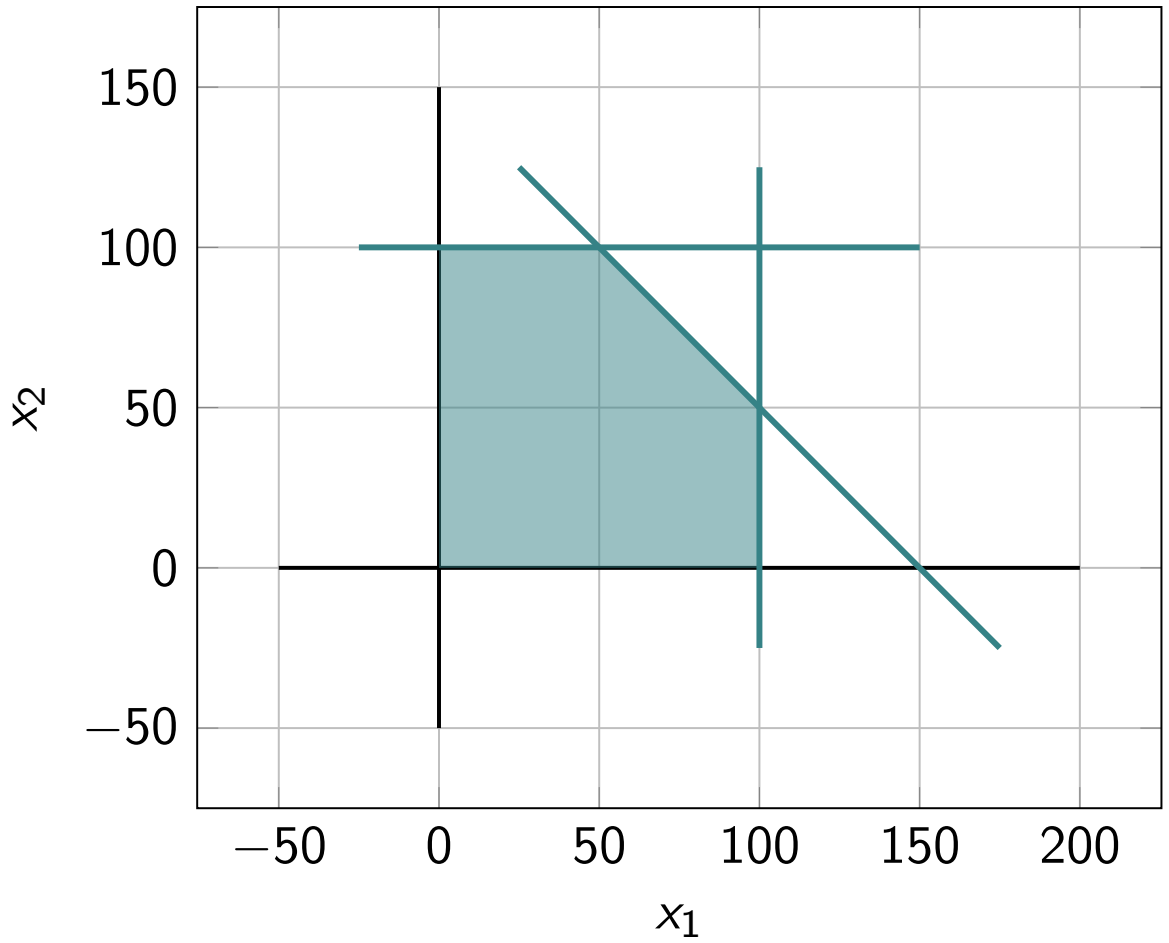
What do we mean by "corner"? Let's formalize this notion.

Definition: Extreme Point

Let P be a polyhedron. A point $\boldsymbol{x} \in P$ is said to be an extreme point of P if we cannot find two vectors $\boldsymbol{y}, \boldsymbol{z} \in P$ with $\boldsymbol{y}, \boldsymbol{z} \neq \boldsymbol{x}$ and a scalar $\lambda \in [0, 1]$, such that $\boldsymbol{x} = \lambda \boldsymbol{y} + (1 - \lambda)\boldsymbol{z}$.

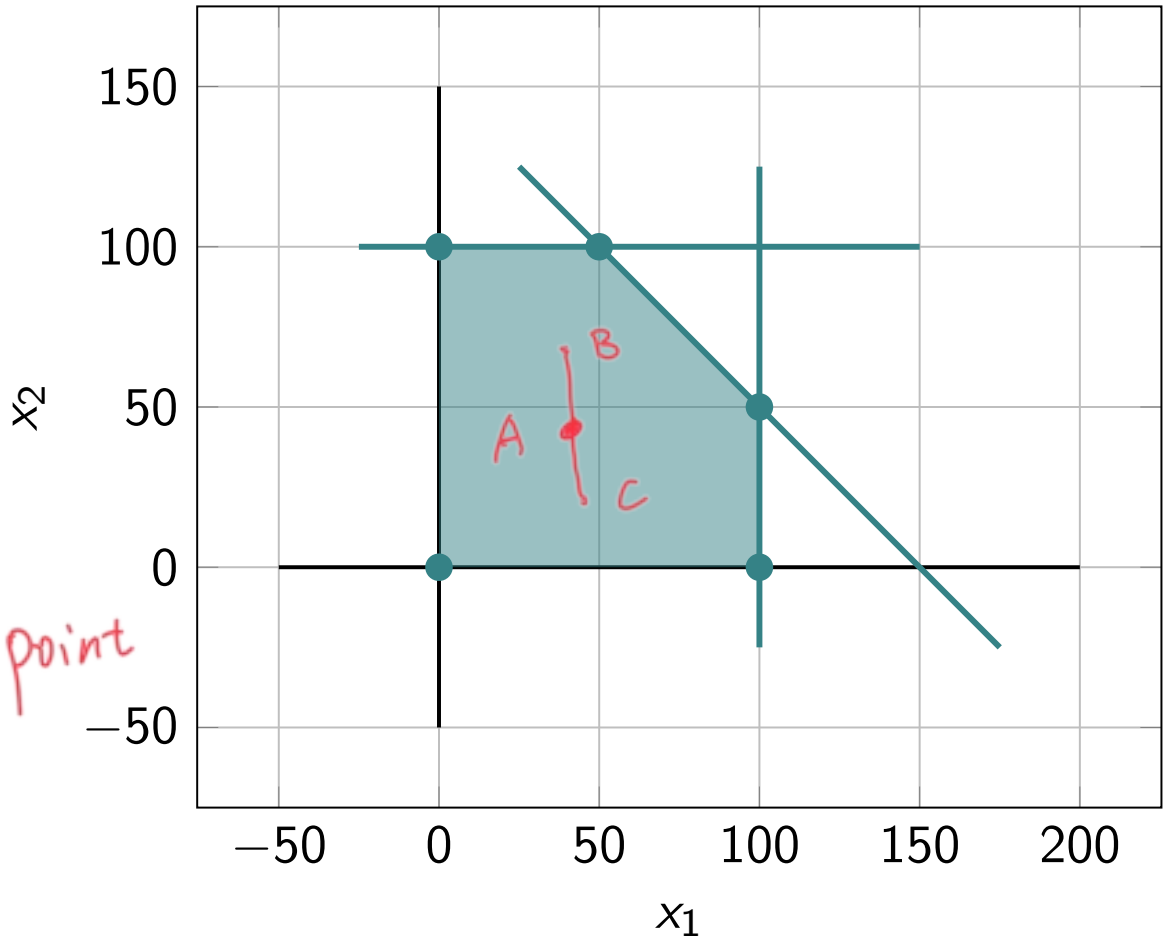
- That is, $\boldsymbol{x}$ cannot be represented as a convex combination of other two points in P.
- We sometimes call the extreme point the vertex or corner of the polyhedron.

Example: Extreme Points



How many extreme points are there in this feasible region?

Example: Extreme Points



A is not an extreme point

How many extreme points are there in this feasible region?

- **Answer:** 5

Extreme Points

We just introduced the definition of extreme points/vertices.

However, this does not tell us how to find those points. We want to have a good way for finding extreme points.

Preview of the Next Steps:

- We need an algebraic way to represent extreme points.
- We will show that it is sufficient to look at extreme points to solve a linear optimization problem.
- Finally, in order find the optimal extreme point, this will lead to the construction of the simplex algorithm for solving LPs.

Basic Solutions

Finding Extreme Points of an LP

In the following, we consider an LP in its standard form:

$$\begin{aligned} \text{minimize} \quad & \boldsymbol{c}^\top \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\ & \boldsymbol{x} \geq \boldsymbol{0}, \end{aligned}$$

where $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{A}$ is an $m \times n$ matrix ($m < n$) and $\boldsymbol{b} \in \mathbb{R}^m$.

General Assumption:

$\boldsymbol{A}$ has linearly independent rows (or equivalently $\boldsymbol{A}$ has rank m).

What happens if this condition is not satisfied?

- Then either there is a redundant constraints (in which case one can remove it) or the constraints are not consistent (in which case there is no feasible point).

Extreme Points of LPs: Basic Solutions

Now, we study the extreme points of an LP in their algebraic form.

Definition: Basic Solution

We call $\boldsymbol{x}$ a basic solution of the LP if and only if

1. $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$.
2. There exist indices $B(1), \ldots, B(m)$ such that the columns of

$$\begin{bmatrix} | & | & & | \\ \boldsymbol{A}_{B(1)} & \boldsymbol{A}_{B(2)} & \cdots & \boldsymbol{A}_{B(m)} \\ | & | & & | \end{bmatrix}$$

are linearly independent and $x_i = 0$ for $i \neq B(1), \ldots, B(m)$.

Finding a Basic Solution

We write

$$\boldsymbol{A}_B = \begin{bmatrix} | & | & & | \\ \boldsymbol{A}_{B(1)} & \boldsymbol{A}_{B(2)} & \cdots & \boldsymbol{A}_{B(m)} \\ | & | & & | \end{bmatrix}, \quad \boldsymbol{x}_B = \begin{bmatrix} x_{B(1)} \\ \vdots \\ x_{B(m)} \end{bmatrix}.$$

Procedure to Find a Basic Solution:

1. Choose any m independent columns of $\boldsymbol{A}$: $\boldsymbol{A}_{B(1)}, \ldots, \boldsymbol{A}_{B(m)}$.
2. Let $x_i = 0$ for all $i \neq B(1), \ldots, B(m)$.
3. Solve the equation $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ for the remaining $x_{B(1)}, \ldots, x_{B(m)}$. This is equivalent to $\boldsymbol{A}_B \boldsymbol{x}_B = \boldsymbol{b}$ or $\boldsymbol{x}_B = \boldsymbol{A}_B^{-1} \boldsymbol{b}$.

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- Since $\boldsymbol{A}_{B(1)}, \ldots, \boldsymbol{A}_{B(m)}$ are linearly independent, $\boldsymbol{A}_B$ is invertible and the last step has a unique solution.
- Basic solution of an LP only depends on its constraint $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$.

Basic Solutions: Notation

$$\boldsymbol{A}_B = \begin{bmatrix} | & | & & | \\ \boldsymbol{A}_{B(1)} & \boldsymbol{A}_{B(2)} & \cdots & \boldsymbol{A}_{B(m)} \\ | & | & & | \end{bmatrix}, \quad \boldsymbol{x}_B = \begin{bmatrix} x_{B(1)} \\ \vdots \\ x_{B(m)} \end{bmatrix}.$$

- We call $B = \{B(1), \ldots, B(m)\}$ the basic indices for this basic solution, $\boldsymbol{A}_{B(1)}, \ldots, \boldsymbol{A}_{B(m)}$ the basic columns, $\boldsymbol{A}_B$ the basis matrix and $x_{B(1)}, \ldots, x_{B(m)}$ the basic variables.
- We call the remaining indices the non-basic indices, the remaining columns of $\boldsymbol{A}$ the non-basic columns and the remaining variables the non-basic variables.

Example: Production Planning

Recall the production problem:

$$
\begin{array}{rrrrl}
\text{maximize} & x_1 & +2x_2 & & \\
\text{subject to} & x_1 & & \leq 100 \\
& & 2x_2 & \leq 200 \\
& x_1 & +x_2 & \leq 150 \\
& x_1, & x_2 & \geq 0
\end{array}
$$

with standard form:

$$
\begin{array}{rrrrrrl}
\text{minimize} & -x_1 & -2x_2 & & & & \\
\text{subject to} & x_1 & & +s_1 & & & = 100 \\
& & 2x_2 & & +s_2 & & = 200 \\
& x_1 & +x_2 & & & +s_3 & = 150 \\
& x_1, & x_2, & s_1, & s_2, & s_3 & \geq 0
\end{array}
$$

Example: Continued

We can denote the feasible set by $\{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq \boldsymbol{0}\}$, where

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix} \quad \boldsymbol{b} = \begin{bmatrix} 100 \\ 200 \\ 150 \end{bmatrix}.$$

- Choose 3 independent columns of $\boldsymbol{A}$, e.g., the first 3 ($B = \{1, 2, 3\}$), we get the corresponding basic solution via:

$$\boldsymbol{x}_B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 100 \\ 200 \\ 150 \end{bmatrix} = \begin{bmatrix} 50 \\ 100 \\ 50 \end{bmatrix}.$$

- That is $x_1 = 50, x_2 = 100, s_1 = 50$. Hence, $(50, 100, 50, 0, 0)$ is a basic solution. One can find other basic solutions by choosing other sets of columns.

Example Continued

We can list all basic solutions:

Indices	$\{1,2,3\}$ (A)	$\{1,2,4\}$ (B)	$\{1,2,5\}$ (C)
Solution	(50, 100, 50, 0, 0)	(100, 50, 0, 100, 0)	(100, 100, 0, 0, -50)
Status	feasible	feasible	not feasible

Indices	$\{1,3,4\}$ (D)	$\{1,4,5\}$ (E)	$\{2,3,4\}$ (F)
Solution	(150, 0, -50, 200, 0)	(100, 0, 0, 200, 50)	(0, 150, 100, -100, 0)
Status	not feasible	feasible	not feasible

Indices	$\{2,3,5\}$ (G)	$\{3,4,5\}$ (H)	
Solution	(0, 100, 100, 0, 50)	(0, 0, 100, 200, 150)	
Status	feasible	feasible	

The other two choices $\{1,3,5\}$ and $\{2,4,5\}$ lead to dependent basic columns (therefore no basic solutions can be obtained).

- Not all BS are feasible, because $\boldsymbol{x}_B$ can have negative coordinates.

Question

How many non-zeros could one have in a basic solution (assuming there are m constraints)?

- No more than m.
- Could be anything between 0 to m (typically it is m).

How many basic solutions can one have for a linear program with m constraints and n variables?

- At most $C(n, m) = \frac{n!}{m!(n-m)!}$ (Combination number).
- Therefore for a finite number of linear constraints, there can only be a finite number of basic solutions!

Basic Feasible Solutions

Basic Feasible Solutions

Definition: Basic Feasible Solution

If a basic solution $\boldsymbol{x}$ also satisfies that $\boldsymbol{x} \geq \mathbf{0}$, then we call it a basic feasible solution (BFS).

How to Find a BFS?

- First find a basic solution $\boldsymbol{x}$.
- Check if $\boldsymbol{x} \geq \mathbf{0}$.

Theorem: Extreme Points and BFS

For the standard LP polyhedron $P := \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \boldsymbol{x} \geq \mathbf{0}\}$, the following statements are equivalent:

1. $\boldsymbol{x}$ is an extreme point of P.
2. $\boldsymbol{x}$ is a basic feasible solution.

Back to the Example

We can list all basic solutions:

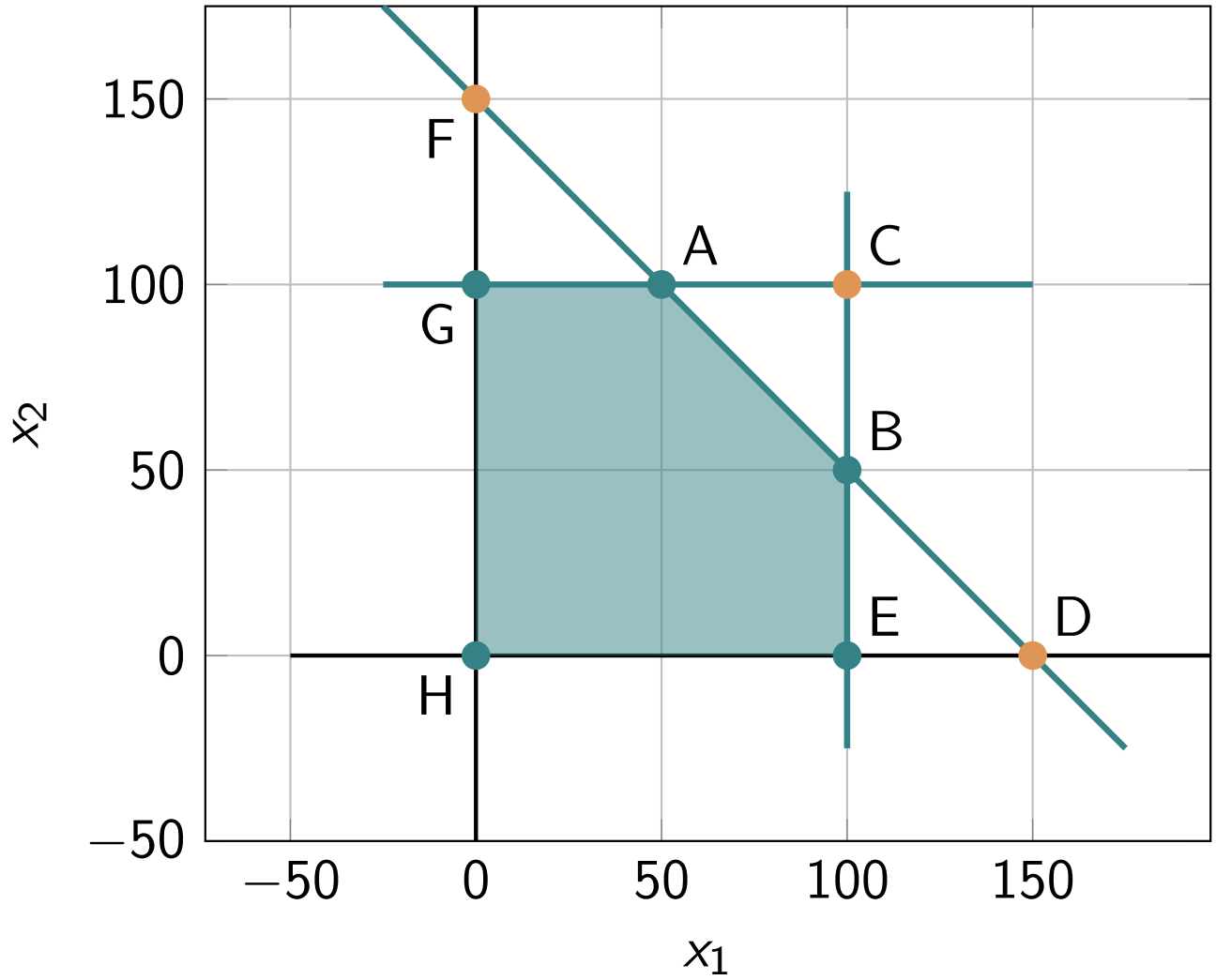
Indices	$\{1,2,3\}$ (A)	$\{1,2,4\}$ (B)	$\{1,2,5\}$ (C)
Solution	(50, 100, 50, 0, 0)	(100, 50, 0, 100, 0)	(100, 100, 0, 0, -50)
Status	feasible	feasible	not feasible

Indices	$\{1,3,4\}$ (D)	$\{1,4,5\}$ (E)	$\{2,3,4\}$ (F)
Solution	(150, 0, -50, 200, 0)	(100, 0, 0, 200, 50)	(0, 150, 100, -100, 0)
Status	not feasible	feasible	not feasible

Indices	$\{2,3,5\}$ (G)	$\{3,4,5\}$ (H)	
Solution	(0, 100, 100, 0, 50)	(0, 0, 100, 200, 150)	
Status	feasible	feasible	

Back to the Example

The BFSs A, B, E, G, H correspond to all extreme points.



Proof Sketch

We show how to prove $1 \Rightarrow 2$ (the other part can be found in Chapter 2 of Bertsimas's book). First, if $\mathbf{x}$ is an extreme point, then it must be a feasible solution. In the following, we prove by contradiction that $\mathbf{x}$ must be a basic solution as well.

Let $B = \{B(1), ..., B(k)\}$ be the set of indices such that $x_i > 0$. If $\mathbf{x}$ is not a basic solution, then it must be that Columns $A_{B(1)}, ..., A_{B(k)}$ are linearly dependent. (Why?)

Then there must exists $(\alpha_1, ..., \alpha_k) \neq 0$ such that $\alpha_1 A_{B(1)} + \cdots + \alpha_k A_{B(k)} = 0$. Since $x_i > 0$ for all $i \in B$, we can choose a small ϵ such that we can construct $\bar{\boldsymbol{x}}$ and $\tilde{\boldsymbol{x}}$

$$\bar{x}_i = \begin{cases} x_i + \epsilon\alpha_j & i = B(j) \\ 0 & i \notin B \end{cases} \quad \text{and} \quad \tilde{x}_j = \begin{cases} x_i - \epsilon\alpha_j & i = B(j) \\ 0 & i \notin B \end{cases}$$

and $\bar{\boldsymbol{x}}$ and $\tilde{\boldsymbol{x}}$ are feasible. (Why?)

Then we can verify $\mathbf{x} = 0.5\bar{\mathbf{x}} + 0.5\tilde{\mathbf{x}}$, which contradicts with that $\mathbf{x}$ is an extreme point. □

Today's lecture

- Sove LP from graph
- Definition of extreme point
- Definition of BS and BFS
- Equivalency between extreme point and BFS in LP

Next lecture

- Fundamental theorem of LP
- Simplex method

Questions?